

Mahima

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Researcher focused on software sustainability and cloud computing. Currently developing metrics to measure the carbon intensity of cloud-based software systems. Passionate about green tech solutions and reducing the environmental impact of software.

Research Experience

Researcher
Aug 2024 – Present

International Institute of Information Technology, Bangalore

- Developing metrics to compute the carbon intensity of applications in hybrid cloud environments.
- Analyzing the energy consumption various cloud architectures like serverless applications, applications with provisioned server, containerized workloads, etc.

Experience

Member Technical
Apr 2021 – May 2024, Hyderabad, IN

ADP India Private Limited

- Designed and developed custom reports** for the Lifion Human Capital Management (HCM) system, providing actionable insights to stakeholders. Utilized advanced data analysis techniques to ensure compliance and accuracy in large-scale payroll systems
- Built and optimized custom data integrations** for Recruitment Management systems, improving process efficiency and enabling seamless data exchange.
- Conducted quality assurance testing** to validate data accuracy, report functionality, and integration performance, ensuring adherence to business requirements.
- Created and optimized SQL queries** for data extraction, analysis, and reporting, streamlining workflows and improving database performance.
- Mentored a team member** through the complete project lifecycle, providing guidance from initiation to delivery.
- Delivered comprehensive training sessions** on project workflows and product usage for new joiners to accelerate their project onboarding.

Trainee Software Developer
Jun 2020 – Dec 2020, Mangalore, IN

Winman Software India LLP

- Developed SQL queries for backend database functionality, ensuring application reliability.
- Conducted unit testing for backend applications to improve system performance.

Education

Master of Science by Research - Computer Science
International Institute of Information Technology, Bangalore
Expected Jul 2026
Research Focus: Developing Metrics to Measure Software Carbon Intensity in Cloud Applications
Cumulative GPA(Current): 3.36 / 4.0

Bachelor of Engineering - Computer Science and Engineering
Visvesvaraya Technological University, Belgaum
Aug 2016 – Aug 2020
Cumulative GPA: 7.8 / 10

Skills

Programming Languages: Python, Java, SQL

Tools & Technologies: Git, Docker, Kubernetes, Jenkins, AWS, Linux

Machine Learning: Pandas, Numpy, Scikit-learn

Cloud Platforms: Amazon Web Services

Projects

Home Security Application (Serverless | AWS) *AWS (Lambda, S3, DynamoDB, API Gateway, Rekognition), Python, Node.js, Java, HTML, CSS, JS*

- Designed and developed a serverless smart door access system simulating real-world home security workflows.
- Implemented facial recognition using Amazon Rekognition to authenticate users before granting access.
- Integrated PIN-based secondary authentication for enhanced security.
- Built an event-driven backend using AWS Lambda, API Gateway, DynamoDB, and Amazon S3 for image storage.
- Developed frontend using HTML, CSS, and JavaScript.
- Used Python, Node.js, and Java for backend services and integrations.
- Achieved a highly scalable, low-maintenance serverless architecture with minimal operational overhead.

Pressure Temperature Monitoring System (Cloud, IoT Containers) *AWS (EC2), Node.js, HTML, CSS, JS, HTTP, MQTT, Docker, Kubernetes*

- Developed a real-time pressure and temperature monitoring system using simulated sensor data.
- Implemented threshold-based logic to automatically actuate valves.
- Built and deployed multiple architectural versions:
 - HTTP-based application hosted on AWS EC2.
 - MQTT-based version for efficient IoT communication.
 - Containerized microservice using Docker and Kubernetes.
- Designed backend services using Node.js with a web-based monitoring interface.
- Deployed and managed applications on AWS EC2, gaining hands-on experience with cloud infrastructure and container orchestration.

Mutation Testing on Library Management System

Java, PIT Tool

- Conducted mutation testing on a Library Management System to evaluate and improve the test suite quality.
- Achieved a mutation score of 64%, improving code coverage and identifying hidden defects.

Classification Models for Healthcare Test Results

[Random Forest, SVM, Neural Networks]

- Developed machine learning models to classify healthcare test results (Abnormal, Normal, Inconclusive).
- Implemented preprocessing pipelines and achieved 85% model accuracy.

DevOps Automation for Scientific Calculator *[Java, Docker, Jenkins, Git, Kubernetes, Ansible]*

- Developed a scientific calculator in Java, with automated CI/CD pipelines using Jenkins.
- Containerized the app using Docker and Kubernetes, and managed deployments with Ansible.

Certifications & Workshops

Bootcamp on End-to-End AI Lifecycle

ACM India, IBM Research India, IIIT Bangalore

May 2025 - Participated in a hands-on bootcamp covering the full AI lifecycle, including data collection, preprocessing, model development, deployment, and monitoring. - Gained practical insights into AI/ML lifecycle stages, such as model training and evaluation, and deployment for real-world use cases. - Collaborated with industry experts to learn about trust in AI, focusing on safety, reliability, and transparency in AI systems.

Summer School on IoT and Next-Generation Networks

ACM

Jun 2025 - Attended a summer school focusing on IoT protocols, sensors, actuators, and next-gen network technologies. - Explored applications of AI/ML in IoT, enhancing understanding of how these technologies are revolutionizing communication systems.

Workshop on Computer Systems and Machine Learning

Indian Institute of Science (IISc), Bangalore

November 2025 - Participated in a workshop focused on the integration of machine learning applications with modern computing systems. - Learned about the challenges in creating energy-efficient, secure, and reliable systems for machine learning applications, like deep neural networks. - Interacted with researchers and professionals to understand future computing systems and advancements in machine learning algorithms.